VTPEH 6270 - Check Point 02 - Key

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1 Introduction

1.1 Background

Toxics Release Inventory (TRI) data track how industrial facilities manage and release toxic chemicals to the environment. These data can support environmental health monitoring by identifying where toxic releases are concentrated and which industries or chemicals contribute most. Understanding spatial and sectoral patterns of releases can help generate hypotheses about environmental exposure and potential inequities.

1.2 Project question

How do total reported toxic chemical releases vary across U.S. states in 2024, and which states report the highest total releases?

1.3 Dataset overview

This project uses the EPA Toxics Release Inventory (TRI) Basic Data File for 2024 (U.S.). The dataset contains facility-reported records describing toxic chemicals, facility location (state/county), industry sector, and quantities released or managed (e.g., total releases and air/water release components). These variables allow comparisons of release amounts across locations and sectors.

1.4 Dataset link

EPA Toxics Release Inventory (TRI) Basic Data Files (1987–present): https://www.epa.gov/toxics-release-inventory-tri-program/tri-basic-data-files-calendar-years-1987-present

2 Data Set Exploration

2.1 Import data

```r
# Packages
library(ggplot2)

# Set working directory
setwd("C:/Users/rui/Desktop/VTPEH 6270/2.5/Data")

# Load TRI data
tri <- read.csv("tri_2024_us.csv", check.names = FALSE)

# Quick check
dim(tri)
names(tri)[1:20]
head(tri[, 1:8])
```

2.2 Keep data tidy

The TRI dataset is already in a tidy tabular structure where each row represents a reported record and columns represent variables (location, industry, chemical, and release quantities). To keep the workflow manageable, I created a subset containing no more than 10 variables.

2.3 Create a subset with <=10 variables

```r
#Subset variables (<=10 variables)
tri_sub <- tri[, c(
  "1. YEAR",                    # categorical (year)
  "8. ST",                      # categorical (state)
  "7. COUNTY",                  # categorical (county)
  "23. INDUSTRY SECTOR",        # categorical (industry sector)
  "37. CHEMICAL",               # categorical (chemical name)
  "107. TOTAL RELEASES",        # numeric (lbs)
  "51. 5.1 - FUGITIVE AIR",     # numeric (lbs)
  "52. 5.2 - STACK AIR",        # numeric (lbs)
  "53. 5.3 - WATER",            # numeric (lbs)
  "65. ON-SITE RELEASE TOTAL"   # numeric (lbs)
)]

#clean variable names
names(tri_sub) <- c(
  "year", "state", "county", "industry_sector", "chemical",
  "total_releases", "fugitive_air", "stack_air", "water", "onsite_total"
)
```

2.4 Variable summary table

```r
var_table <- data.frame(
  Variable = c(
    "year", "state", "county", "industry_sector", "chemical",
    "total_releases", "fugitive_air", "stack_air", "water", "onsite_total"
  ),
  Description = c(
    "Reporting year",
    "U.S. state where facility is located",
    "County where facility is located",
    "Industrial sector of the reporting facility",
```

```r
    "Name of the reported chemical",
    "Total quantity of chemical releases (lbs)",
    "Fugitive air emissions (lbs)",
    "Stack air emissions (lbs)",
    "Water discharges (lbs)",
    "Total on-site releases (lbs)"
  ),
  Type = c(
    "Discrete", "Categorical", "Categorical", "Categorical", "Categorical",
    "Continuous", "Continuous", "Continuous", "Continuous", "Continuous"
  ),
  Class = sapply(tri_sub, class)
)
# Generate a summary table of variables for reporting
knitr::kable(
  var_table,
  caption = "Table 1. Overview of variables included in the TRI 2024 subset"
)
```

3 Data Visualization

3.1 Histogram

Total toxic chemical releases are highly right-skewed; therefore, a log(1+x) transformation was applied to improve visualization while retaining facilities with zero reported releases.

```r
# Create log-transformed variable (handles zeros safely)
tri_sub$log_total_releases <- log1p(tri_sub$total_releases)

# Compute mean and median on the transformed scale
mean_log <- mean(tri_sub$log_total_releases, na.rm = TRUE)
median_log <- median(tri_sub$log_total_releases, na.rm = TRUE)

# Plot histogram with mean/median lines and legend
ggplot(tri_sub, aes(x = log_total_releases)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  geom_vline(aes(xintercept = mean_log, color = "Mean"),
             linewidth = 1.2, linetype = "dashed") +
  geom_vline(aes(xintercept = median_log, color = "Median"),
             linewidth = 1.2) +
  scale_color_manual(values = c("Mean" = "red", "Median" = "darkgreen")) +
  scale_x_continuous(expand = c(0, 0)) +
  scale_y_continuous(expand = c(0, 0)) +
  labs(
    title = "Distribution of Total Toxic Chemical Releases (log1p)",
    x = "log(1 + total releases in pounds)",
    y = "Number of facilities",
    color = "Statistic",
    caption = "Figure 1. Histogram of total toxic chemical releases in 2024."
  ) +
  theme_classic()
```

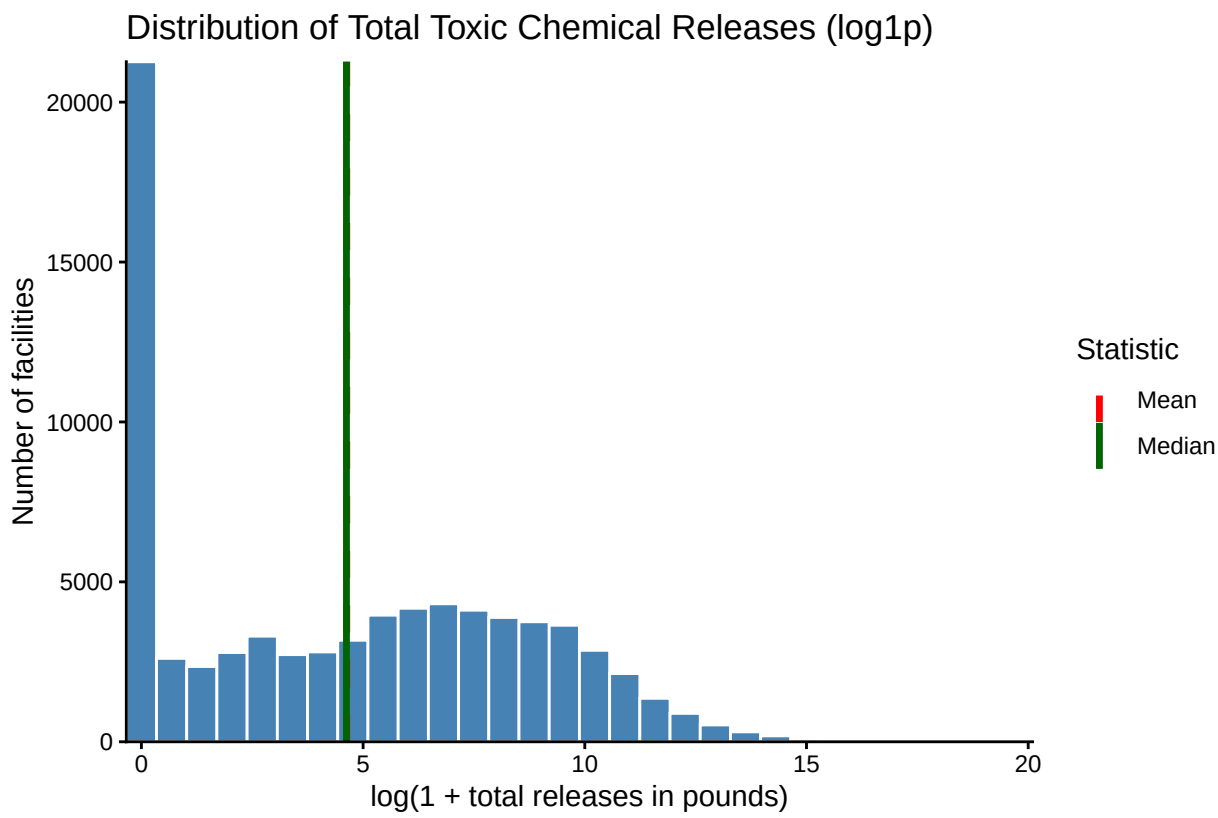

Figure 1. Histogram of total toxic chemical releases in 2024.

The distribution of total toxic chemical releases is highly right-skewed, with most facilities reporting low release amounts and a small number reporting extremely high values.

3.2 Scatterplot Chosen variables: fugitive_air vs stack_air

```
ggplot(tri_sub, aes(x = fugitive_air, y = stack_air)) +
  geom_point(alpha = 0.4, color = "darkorange") +
  labs(
    title = "Fugitive vs. Stack Air Releases",
    x = "Fugitive air releases (pounds)",
    y = "Stack air releases (pounds)",
    caption = "Figure 2. Scatterplot comparing fugitive and stack air toxic releases in 2024."
  ) +
  scale_x_continuous(labels = scales::comma) +
  scale_y_continuous(labels = scales::comma) +
  theme_classic()
```

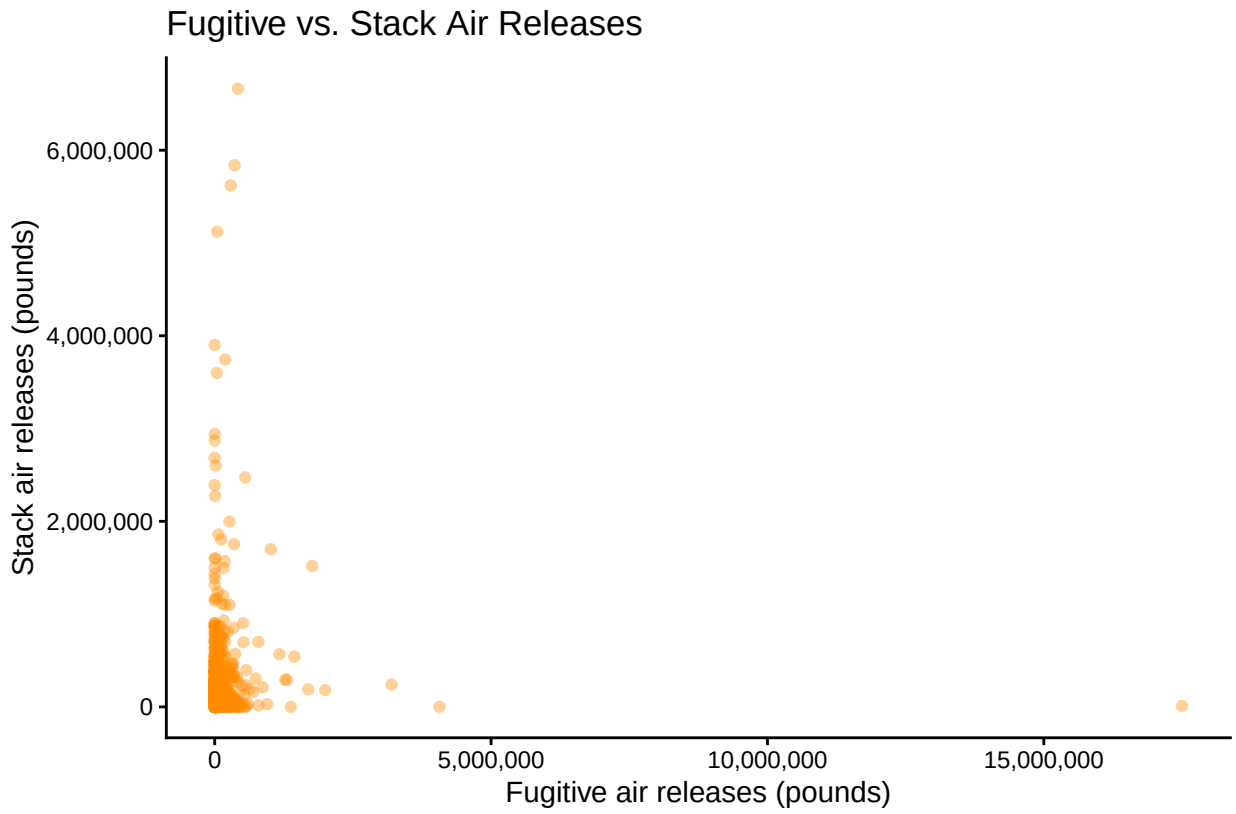

Figure 2. Scatterplot comparing fugitive and stack air toxic releases in 2024.

There is substantial variability in air release pathways, with no clear linear relationship between fugitive and stack air releases across facilities.

4 AI Use Disclosure Statement

This document was generated with assistance from ChatGPT, which was used to help understand the assignment requirements and to debug and improve R code written by the author. All analyses, interpretations,and conclusions were made by the author.